ADVANCED TECHNOLOGICAL INSTITUTE

Higher National Diploma in Information Technology

Project Proposal

GROUP PROJECT

Year:

1st Year - 2nd Semester

Supervisor:

Mrs. Priyangika Aththanayake

Project:

Tuition Class Website (Sigma Educational Institute)

Group No:

03

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INTRODUCTION

A Course Management System is a software platform designed to facilitate the organization, delivery, and management of educational courses. It provides a centralized digital environment where educators can create and distribute course materials, assign tasks, track student progress, and interact with students. Common features of a CMS include content uploading (such as lectures, readings, and multimedia), assignment submission, grading tools, discussion forums, and calendars for scheduling. These systems are widely used in educational institutions and corporate training programs to enhance teaching efficiency and support remote or blended learning.

Sigma Institute, a reputed educational institution, currently operates without an official website, limiting its ability to communicate effectively, share essential information, and enhance its institutional image in the digital space. In today's technology-driven world, an online presence is no longer optional but a necessity for academic institutions to connect with students, parents, educators, and stakeholders. And also it is difficult to maintain consistent communication, distribute study materials and provide learning support online. Managing students, their progress, and interactions in Sigma Institute is often done manually. Because of this Sigma Institute often face challenges in maintaining student records, tracking performance, and facilitating communication. A lack of an integrated system can lead to inefficiency and mismanagement. Recognizing this critical need, this project aims to bridge that gap by developing a comprehensive and user-friendly web application for Sigma Institute, that acts as a central hub for students and teachers.

With our web application, students can access classes online from anywhere, eliminating the limitation of physical centers. This makes learning more inclusive and convenient. The new platform will serve as a single system to manage student records, course enrollments, schedules and academic progress, ensuring consistency and efficiency. Moreover, our web application will include video lectures, interactive study materials, and online assessments making education more engaging and effective with features like chat support and notifications, students and teachers can communicate seamlessly, enhancing engagement. A built analytics system will allow tracking of student progress, performance, feedback collection, ensuring continuous improvement in learning outcomes. Tasks like student registration, course enrollment, fee payments and communication will be automated, reducing manual work and improving operational efficiency.

Business Process

Sigma Institute, prior to the implementation of a Course Management System, operates largely through manual processes, which involve high levels of paperwork, inefficiencies, and fragmented communication. The proposed digital platform seeks to transform these outdated practices into streamlined, centralized digital workflows.

The **admission and registration process** begins when prospective students apply for enrollment. Traditionally, this has been a paper-based process where forms are physically submitted and verified by administrative staff. With the CMS in place, this entire procedure will shift online. Students will complete digital registration forms, upload necessary documents, and receive real-time updates on their application status. The system will automatically generate student IDs and maintain accurate, searchable records, reducing the workload on staff and minimizing errors.

Once registered, students will move to **course enrollment and scheduling**. Previously, course assignments and timetables were manually created and shared via physical notice boards or in-person announcements. The new system allows students to log in, view available courses, and enroll in them digitally. Timetables will be auto-generated and dynamically updated, ensuring students and teachers always have access to the latest schedules. Notification features will remind users of upcoming classes, changes, or cancellations.

The **teaching and learning process** will also be significantly enhanced. Traditionally, instruction was delivered solely through in-person classes, with limited support for remote learners. Teachers distributed notes manually, and students often lacked access to centralized learning materials. The CMS will provide a dedicated portal where teachers can upload video lectures, reading materials, presentations, and interactive content. This enables asynchronous learning and ensures that all students, regardless of location, can access high-quality resources at any time.

Assignment submission and evaluation have historically been labor-intensive. Students submitted physical copies of assignments, and teachers manually graded and returned them. These documents were easily misplaced, and feedback was often delayed. Through the CMS, students will submit assignments online within specified deadlines. Teachers can review and grade work directly on the platform, provide timely feedback, and track submissions efficiently. Grading rubrics and auto-grading features for objective questions will further streamline evaluation.

The **monitoring of student performance** is another area seeing major improvement. Previously, performance tracking involved maintaining multiple paper records, making it difficult to get a comprehensive view of student progress. With the CMS, every grade, attendance record, and activity is logged automatically. Teachers and administrators can generate analytics reports, identify struggling students early, and provide targeted academic support. Students will also have access to personal dashboards where they can monitor their progress over time.

Communication and support mechanisms were previously informal and inconsistent, relying heavily on word-of-mouth, handwritten notices, or phone calls. The new platform integrates chat systems, message boards, and announcements, enabling smooth, documented communication between students, teachers, and administrators. Real-time alerts ensure no critical updates are missed.

Another crucial aspect is **fee management**, which was traditionally handled via cash or manual banking procedures. Receipts were handwritten, and tracking fee payments was error-prone. The CMS introduces a secure online payment gateway where students can pay tuition and other fees, receive digital receipts instantly, and view payment history. Automated reminders for due payments and penalty calculations ensure better financial tracking.

Finally, **feedback collection and continuous improvement** are embedded into the system. Previously, collecting student feedback was rare or conducted via paper surveys with limited follow-up. With the CMS, periodic digital feedback forms will be deployed automatically after course completion. Responses will be analyzed to generate actionable insights, helping administrators and educators refine course content, teaching methods, and overall service quality.

Aims & Objectives

Goals:

1. Improve Operational Efficiency Through Digitization
2. Enhance Teaching and Learning Experiences
3. Strengthen Student Monitoring and Academic Performance
4. Improve Communication and Engagement
5. Ensure Transparent and Secure Financial Transactions
6. Support Institutional Growth and Quality Improvement

Objectives:

1. Automate student admission and registration processes to reduce paperwork and administrative workload. Digitize class schedules, course management, and faculty assignments to centralize and streamline academic planning. Minimize manual errors in student data handling by implementing a secure, centralized database.

2. Provide an online platform for faculty to upload lectures, reading materials, and interactive content. Enable students to access course content anytime, anywhere, supporting flexible and remote learning. Facilitate online assessments and assignments with timely grading and feedback.

3. Track student attendance, grades, and submission records in real time. Generate performance analytics and reports to identify students needing academic support. Maintain academic history and performance trends for each student to support long-term planning and counseling.

4. Implement an in-platform chat, messaging, and notification system for seamless communication. Ensure timely announcements and class updates reach all stakeholders without delays. Foster collaboration between students and teachers via discussion forums and message boards.

5. Integrate an online payment gateway for tuition and fee processing. Automate payment tracking, receipt generation, and pending fee alerts. Allow students and staff to view detailed fee histories and generate reports.
6. Collect structured feedback from students about courses and faculty performance. Use feedback data to make informed decisions on course improvements and faculty training. Build a modular and scalable system that can adapt to future academic programs and student growth.

Scope of Work

1. User Roles & Authentication

- **Admin:**
  Manages all users (students, teachers), class content, system settings, and overall platform operations.
- **Teachers:**
  Create and manage classes, upload study materials, evaluate student performance, and communicate with learners.
- **Students:**
  Register and enroll in classes, access content, complete assignments, and take quizzes.
- **Secure Login System:**
  Role-based login via email/password and optional Google authentication.

2. Class Management

- Teachers can create, edit, and publish classes.
- Support for multiple content formats such as videos, PDFs, notes, and interactive quizzes.
- Class categorization based on subjects, difficulty levels, or departments.
- Access control for managing class availability and student enrollment.

3. Student Learning & Progress Tracking

- Real-time dashboards to monitor student engagement and learning progress.
- Detailed performance reports available to teachers and administrators.

4. Communication & Collaboration

- Built-in messaging system for communication between students and teachers.
- Integration with Zoom or Google Meet for conducting live classes and virtual discussions.

5. Payment & Subscription (Optional)

- Integration of payment gateways (e.g., PayPal, Stripe) for paid classes.
- Subscription models for accessing premium or advanced-level content.

6. Admin Panel

- Full user management: add, edit, or remove students and teachers.
- Approve and moderate classes submitted by instructors.
- Generate analytical reports on user activity, class effectiveness, and platform usage.

7. Security & Performance

- Role-based access control and secure login mechanisms.
- Data encryption to protect sensitive user information.
- Scalable architecture to handle a growing number of users and data.

Target Audience

- **Prospective Students & Parents:** Seeking information on classes and admissions.
- **Current Students:** Accessing portals for academic and administrative support.
- **Educational Partners & Stakeholders:** Exploring collaboration opportunities.

RESOURCE REQUIRMENTS

Software

- Visual Studio Code
- Xampp Server
    - Phpmyadmin
- Web Browser

Programming Language

- HTML
- CSS
- JavaScript
- PHP

PROJECT SHEDULE

Task Name	Weeks												
	1	2	3	4	5	6	7	8	9	10	11	12	13
Create Project Proposal													
Project Proposal Presentation													
Create Project Plan													
Prepare Project Budget													
Project Development													
Create Project Report													
Create Project Blog													
Project Evaluation													

APPROVAL

I hereby certify this project is under my supervision and chosen for the approved list of projects.

2025/04/29 ………………………

Date (Project Supervisor)